

8.012 Problem Set 7

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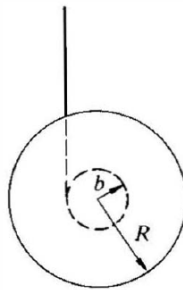
December 2025

Problem 7

Kleppner & Kolenkow, Problem 6.29

6.29 A Yo-Yo of mass M has an axle of radius b and a spool of radius R . Its moment of inertia can be taken to be $MR^2/2$ and the thickness of the string can be neglected. The Yo-Yo is released from rest.

- a. What is the tension in the cord as the Yo-Yo descends and as it ascends?
- b. The center of the Yo-Yo descends distance h before the string is fully unwound. Assuming that it reverses direction with uniform spin velocity, find the average force on the string while the Yo-Yo turns around.



a.

In this problem, the condition of Pure Rolling on the string is

$$a = b\alpha$$

Descends :-

Let a represents the Downward Acceleration
Force Equation :-

$$Ma = Mg - T \implies a = g - \frac{T}{M} \implies \alpha = \frac{g}{b} - \frac{T}{bM}$$

Torque Equation :-

$$\tau = I\alpha \implies Tb = \frac{MR^2}{2}\alpha = \frac{MR^2}{2}\left(\frac{g}{b} - \frac{T}{bM}\right)$$

This gives us

$$T = \frac{MR^2g}{2b^2 + R^2}$$

Ascends :-

Let a represents Upward Acceleration
Force Equation :-

$$Ma = T - Mg \implies a = \frac{T}{M} - g \implies \alpha = \frac{T}{bM} - \frac{g}{b}$$

Torque Equation :-

$$\tau = I\alpha \implies Tb = \frac{MR^2}{2}\alpha = \frac{MR^2}{2}\left(\frac{T}{bM} - \frac{g}{b}\right)$$

This gives us

$$T = \frac{MR^2g}{2b^2 - R^2}$$

b.

At the instant the string is fully unwound, the Yo-Yo has translational speed v and angular speed ω related by

$$v = b\omega.$$

Using conservation of mechanical energy during the descent,

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{MR^2}{2}\right)\left(\frac{v}{b}\right)^2.$$

Hence,

$$v^2 = \frac{4b^2gh}{2b^2 + R^2}.$$

During the turnaround, the angular velocity remains unchanged. Therefore, no net torque acts on the Yo-Yo, and the impulsive force acts only to reverse the linear momentum of the center of mass.

The change in linear momentum is

$$\Delta p = M(-v) - M(v) = -2Mv.$$

Let Δt be the (short) time interval during which the Yo-Yo reverses direction. The impulse–momentum relation gives

$$(F_{\text{string,avg}} - Mg) \Delta t = 2Mv.$$

Thus, the impulse delivered by the string is

$$J = 2Mv$$

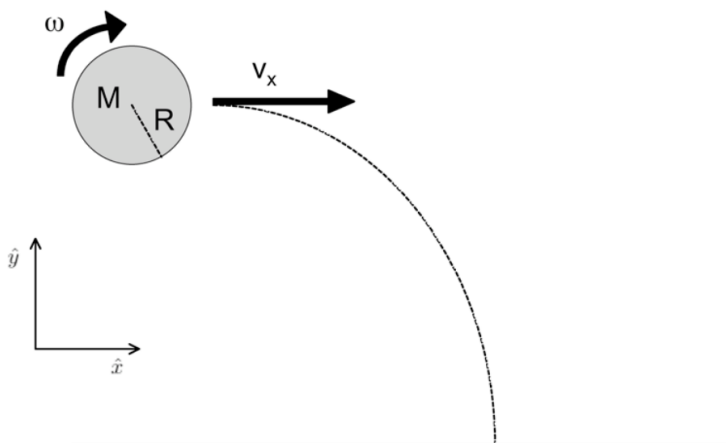
and the average force exerted by the string is

$$F_{\text{string,avg}} = Mg + \frac{2Mv}{\Delta t}$$

where Δt depends on the detailed dynamics of the turnaround and cannot be determined from the given information.

Problem 11

Superball Bounce. [15 points]



A superball of mass M , radius R and uniform density is thrown with horizontal velocity v_x and rotation rate ω (assume that the spin angular momentum vector is orthogonal to the trajectory of the ball). The ball makes a perfectly elastic “bounce” on the floor without slipping, in a manner such that the magnitude of its vertical velocity is the same before and after the bounce (i.e., it would bounce to the same height from which it was originally released).

- [10 pts] Calculate the horizontal velocity and rotation rate of the ball immediately after the bounce, in terms of the v_x , ω and R .
- [5 pts] Calculate the horizontal velocity and rotation rate immediately after a second bounce, in terms of the original v_x , ω and R . Based on this answer, can you describe the motion of the ball after many bounces? (You are encouraged to confirm your answer by experimentation).

a.

During the impulsive collision, external forces act at the point of contact, so angular momentum about the contact point is conserved.

Angular momentum about the contact point just before the bounce:

$$L_{\text{before}} = Mv_x R + I\omega = Mv_x R + \frac{2}{5}MR^2\omega.$$

Just after the bounce:

$$L_{\text{after}} = Mv'_x R + \frac{2}{5}MR^2\omega'.$$

Thus,

$$Mv_x R + \frac{2}{5}MR^2\omega = Mv'_x R + \frac{2}{5}MR^2\omega'. \quad (1)$$

The no-slipping condition immediately after the bounce gives

$$v'_x = \omega' R. \quad (2)$$

Solving (1) and (2), gives us

$$v'_x = \frac{5}{7}v_x + \frac{2}{7}\omega R \quad \omega' = \frac{5}{7}\frac{v_x}{R} + \frac{2}{7}\omega$$

b.

Applying the same transformation again,

$$v''_x = \frac{5}{7}v'_x + \frac{2}{7}\omega' R \quad \omega'' = \frac{5}{7}\frac{v'_x}{R} + \frac{2}{7}\omega'$$

Substituting the results from part (a), we get,

$$v''_x = \frac{25}{49}v_x + \frac{24}{49}\omega R \quad \omega'' = \frac{25}{49}\frac{v_x}{R} + \frac{24}{49}\omega$$

Long-term behavior

During the very short collision with the ground, the external impulsive force acts at the point of contact. Hence, the external torque about the contact point is zero, and angular momentum about this point is conserved.

The angular momentum about the contact point is

$$L = I\omega + MRv.$$

Therefore, conservation of angular momentum gives

$$I\omega^{(n)} + MRv^{(n)} = I\omega^{(n+1)} + MRv^{(n+1)}. \quad (1)$$

At the instant the sphere loses contact with the ground, the no-slip condition must be satisfied:

$$v^{(n+1)} = \omega^{(n+1)} R. \quad (2)$$

Substituting Eq. (2) into Eq. (1),

$$I\omega^{(n)} + MRv^{(n)} = (I + MR^2)\omega^{(n+1)}.$$

Hence,

$$\omega^{(n+1)} = \frac{I\omega^{(n)} + MRv^{(n)}}{I + MR^2}, \quad v^{(n+1)} = R\omega^{(n+1)}. \quad (3)$$

Define the slip variable

$$\Delta^{(n)} \equiv v^{(n)} - \omega^{(n)} R.$$

Using Eq. (3),

$$\Delta^{(n+1)} = v^{(n+1)} - \omega^{(n+1)} R = \frac{I}{I + MR^2} \Delta^{(n)}.$$

For a uniform solid sphere,

$$\frac{I}{I + MR^2} = \frac{\frac{2}{5}MR^2}{\frac{2}{5}MR^2 + MR^2} = \frac{2}{7}.$$

Therefore,

$$\Delta^{(n+1)} = \frac{2}{7} \Delta^{(n)}.$$

Iterating,

$$\Delta^{(n)} = \left(\frac{2}{7}\right)^n \Delta^{(0)}.$$

Since $\left|\frac{2}{7}\right| < 1$, we obtain

$$\lim_{n \rightarrow \infty} \Delta^{(n)} = 0.$$

Hence,

$$\lim_{n \rightarrow \infty} (v^{(n)} - \omega^{(n)} R) = 0$$

which implies that after many bounces the sphere approaches the pure rolling condition

$$v_x^{(n)} \longrightarrow \omega^{(n)} R.$$

so subsequent bounces no longer change the horizontal velocity or angular speed.